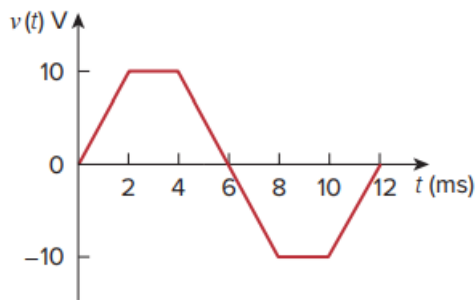


Problem

The voltage waveform in Fig. 6.46 is applied across a $55\text{-}\mu\text{F}$ capacitor. Draw the current waveform through it.

Fig. 6.46:



Step-by-step solution

Step 1 of 9

Refer to Figure 6.46 in the textbook.

Consider the formula for straight line equation for two points (x_1, y_1) and (x_2, y_2) .

$$(y - y_1) = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

Where,

$$y = v(t)$$

$$x = t$$

Step 2 of 9

For $0 \leq t < 2 \text{ ms}$, calculate the straight-line equation of points $(0, 0)$ and $(2\text{m}, 10)$.

$$(y - 0) = \frac{10 - 0}{2 \times 10^{-3} - 0} (x - 0)$$

$$y = \frac{10}{2 \times 10^{-3}} x$$

$$y = 5000x$$

$$v(t) = 5000t \quad \text{for } 0 \leq t < 2 \text{ ms}$$

For $2 \text{ ms} \leq t < 4 \text{ ms}$, the voltage is,

$$v(t) = 10 \quad \text{for } 2 \text{ ms} \leq t < 4 \text{ ms}$$

Step 3 of 9

For $4 \text{ ms} \leq t < 8 \text{ ms}$, calculate the straight-line equation of points $(4\text{m}, 10)$ and $(8\text{m}, -10)$.

$$(y - 10) = \frac{-10 - 10}{8 \times 10^{-3} - 4 \times 10^{-3}} (x - 4 \times 10^{-3})$$

$$y - 10 = -5000(x - 4 \times 10^{-3})$$

$$y - 10 = -5000x + 20$$

$$y = -5000x + 30$$

$$v(t) = -5000t + 30 \quad \text{for } 4 \text{ ms} \leq t < 8 \text{ ms}$$

Step 4 of 9

For $8 \text{ ms} \leq t < 10 \text{ ms}$, the voltage is,

$$v(t) = -10 \quad \text{for } 8 \text{ ms} \leq t < 10 \text{ ms}$$

For $10 \text{ ms} \leq t < 12 \text{ ms}$, calculate the straight-line equation of points $(10\text{m}, -10)$ and $(12\text{m}, 0)$.

$$(y + 10) = \frac{0 + 10}{12 \times 10^{-3} - 10 \times 10^{-3}} (x - 10 \times 10^{-3})$$

$$y + 10 = 5000(x - 10 \times 10^{-3})$$

$$y + 10 = 5000x - 50$$

$$y = 5000x - 60$$

$$v(t) = 5000t - 60 \quad \text{for } 10 \text{ ms} \leq t < 12 \text{ ms}$$

Step 5 of 9

Therefore, the expression for the voltage across the capacitor is,

$$v(t) = \begin{cases} 5000t & \text{for } 0 < t < 2 \\ 10 & \text{for } 2 \text{ ms} < t < 4 \text{ ms} \\ -5000t + 30 & \text{for } 4 \text{ ms} < t < 8 \text{ ms} \\ -10 & \text{for } 8 \text{ ms} < t < 10 \text{ ms} \\ 5000t - 60 & \text{for } 10 \text{ ms} < t < 12 \text{ ms} \end{cases}$$

Step 6 of 9

Write the expression for current flowing through the capacitor.

$$i(t) = C \frac{dv(t)}{dt}$$

For the time interval of $0 < t < 2 \text{ ms}$, the capacitor current is,

$$\begin{aligned} i(t) &= C \frac{dv(t)}{dt} \\ &= 55 \times 10^{-6} \frac{d}{dt}(5000t) \\ &= (55 \times 10^{-6})(5000) \\ &= 0.275 \text{ A} \end{aligned}$$

For the time interval of $2 \text{ ms} < t < 4 \text{ ms}$, the capacitor current is,

$$\begin{aligned} i(t) &= C \frac{dv(t)}{dt} \\ &= 55 \times 10^{-6} \frac{d}{dt}(10) \\ &= 0 \text{ A} \end{aligned}$$

Step 7 of 9

For the time interval of $4 \text{ ms} < t < 8 \text{ ms}$, the capacitor current is,

$$\begin{aligned} i(t) &= C \frac{dv(t)}{dt} \\ &= 55 \times 10^{-6} \frac{d}{dt}(-5000t + 30) \\ &= 55 \times 10^{-6}(-5000) \\ &= -0.275 \text{ A} \end{aligned}$$

For the time interval of $8 \text{ ms} < t < 10 \text{ ms}$, the capacitor current is,

$$\begin{aligned} i(t) &= C \frac{dv(t)}{dt} \\ &= 55 \times 10^{-6} \frac{d}{dt}(-10) \\ &= 0 \text{ A} \end{aligned}$$

Step 8 of 9

For the time interval of $10 \text{ ms} < t < 12 \text{ ms}$, the capacitor current is,

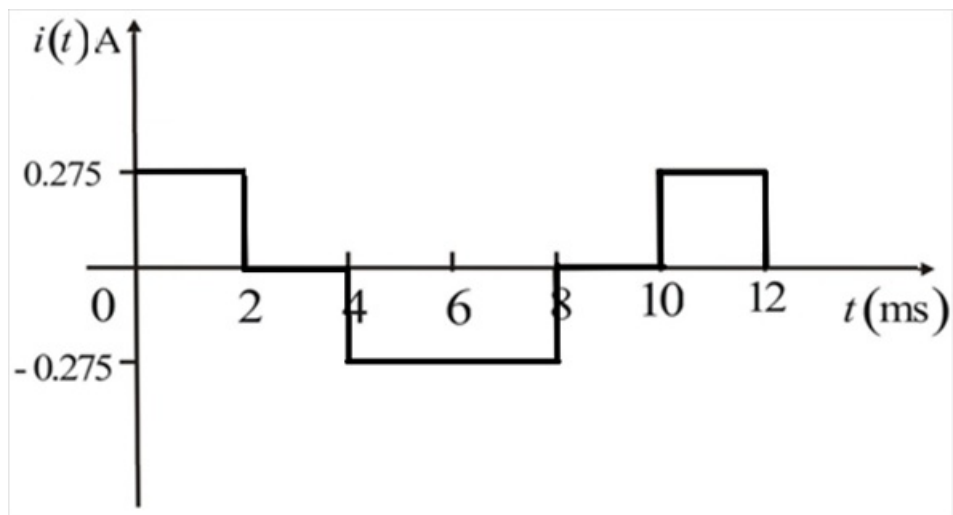
$$\begin{aligned} i(t) &= C \frac{dv(t)}{dt} \\ &= 55 \times 10^{-6} \frac{d}{dt}(5000t - 60) \\ &= 55 \times 10^{-6}(5000) \\ &= 0.275 \text{ A} \end{aligned}$$

Therefore, the current through the capacitor is,

$$i(t) = \begin{cases} 0.275 \text{ A} & \text{for } 0 < t < 2 \\ 0 \text{ A} & \text{for } 2 \text{ ms} < t < 4 \text{ ms} \\ -0.275 \text{ A} & \text{for } 4 \text{ ms} < t < 8 \text{ ms} \\ 0 \text{ A} & \text{for } 8 \text{ ms} < t < 10 \text{ ms} \\ 0.275 \text{ A} & \text{for } 10 \text{ ms} < t < 12 \text{ ms} \end{cases}$$

Step 9 of 9

Draw the waveform of current through the capacitor.



Hence, the required waveform is drawn.